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Sr. No.	Items	Page No.
	Occupation: AI Developer **Occupation content**: Support AI System Development **Research methodology Techniques used**: DACUM (Developing A Curriculum), Competency-Based Training (CBT), and Project-Based Learning approach **Qualification Framework for CBT Program**: NVQ L4 **Unit of Competency**: U01	

1.0 RATIONALE

AI systems are widely used in modern industries. This unit develops basic skills to support AI system development, including data handling and testing. It helps ensure effective and responsible use of AI while improving employability. .

2.0 QUALIFICATION FRAMEWORK FOR CBT PROGRAM

Level of NVQ: Level 04

Sector: ICT

Occupation: AI Developer

Perquisites: G.C.E. Ordinary Level (O/L) with minimum 6 passes including a Credit pass in Mathematics

The CBT program for Support AI Design & Development is structured to develop practical skills and competencies required to design, build, and manage AI-powered support systems. It focuses on hands-on learning, real-world applications, and performance-based assessment.

3.0 COMPETENCY STANDARDS	
Occupation	AI Developer
Unit of Competency	Support AI System Development
Unit Descriptor	This unit describes the competencies required to support the development of AI systems at an entry level. It includes assisting in designing AI solutions, preparing development environments, supporting model training, and integrating AI components under supervision. The unit also covers basic troubleshooting, documentation, and adherence to ethical and data security practices.
Elements of Competency	**Performance Criteria**
1. Assist in designing AI Solutions	**1.1** Problem requirements are identified based on user needs **1.2** Input data and expected outputs are defined clearly **1.3** Appropriate AI method is selected according to the problem **1.4** Basic solution workflow (input → process → output) is designed
2. Support Model Development	**2.1** Development environment is prepared with required tools and libraries **2.2** Dataset is collected and prepared for model training **2.3** Data is cleaned and formatted according to model requirements **2.4** Model training process is executed using given tools and scripts **2.5** Training process is monitored and errors are identified
3. Integrate AI Components	**3.1** Collect, clean, and prepare datasets using appropriate tools and basic AI preprocessing techniques to ensure data quality and usability. **3.2** Select and apply suitable AI algorithms using tools such as Scikit-learn based on the problem requirements and data characteristics. **3.3** Develop and train AI models using programming languages and frameworks such as TensorFlow to produce accurate and reliable outputs. **3.4** Evaluate AI model performance using standard metrics (accuracy, precision, recall) and refine models to improve results. **3.5** Deploy, monitor, and maintain AI solutions while managing versions and updates using tools like Git to ensure continuous system performance.
4. Prepare Development environment	**3.1** Install and configure required software tools such as Python and Visual Studio Code. **3.2** Set up AI libraries and frameworks such as TensorFlow and Scikit-learn. **3.3** Configure virtual environments and manage dependencies for project isolation. **3.4** Set up version control using Git. **3.5** Test and verify the development environment to ensure proper functionality.

Range Statement

1. AI system components (data processing, models, user interfaces)
2. Development tools and frameworks (e.g., Python, TensorFlow)
3. Development activities (coding, testing, debugging)

4.0 COURSE DESIGN	
Course Title	AI Developer
Nominal Duration	120 Hours
Qualification Level	NVQ Level 04
Unit of Competency Support AI System Development	
Course Description	This course develops practical competencies required to support AI system design and development. It covers AI problem analysis, design workflow preparation, development environment setup, data preparation, model support, integration of AI components, and responsible use of AI in practical contexts.
Course Outcomes	• Identify user needs and problem requirements • Define input and output specifications • Select appropriate AI methods based on problem context • Prepare development environments using required tools and frameworks • Assist in developing AI models through data preparation and training support • Integrate AI components into functional systems • Document processes, workflows, and results clearly • Apply ethical standards and data security practices in AI development
Entry Requirement	G.C.E. Ordinary Level (O/L) with minimum 6 passes including a Credit pass in Mathematics.
Assessment Method	Observation, Demonstration, Oral Questioning, Product Evaluation

COURSE STRUCTURE:

Unit of Competency	Module Title	Module Content	Nominal Duration
Support AI System Development	AI Problem Analysis and Solution Design	Identify problem and user requirements; define inputs and outputs; select appropriate AI method; design workflow (input → process → output).	30 Hours
	Prepare Development Environment	Install Python and Visual Studio Code; set up TensorFlow and Scikit-learn; configure virtual environments; manage dependencies; set up Git; verify environment.	25 Hours
	Support Model Development	Collect and prepare dataset; clean and format data; run training scripts; monitor training; identify errors.	35 Hours
	Integrate AI Components	Prepare datasets; apply algorithms; train models; evaluate using accuracy, precision, recall; deploy and maintain with version control.	30 Hours

COMPETENCY ANALYSIS:

Unit of Competency	Module		Industrial Training
Support AI System Development	1	2	3
	AI Problem Analysis and Solution Design	Prepare Development Environment Support Model Development and Integration	Practical exposure in laboratory or workplace to observe and participate in AI-related support activities
Assessment Competency shall be assessed through Observation, Demonstration, Oral Questioning, and Product Evaluation in accordance with CBT guidelines			
Methodology Training shall be delivered through hands-on practice, competency-based training, guided learning, demonstrations, mini-project tasks, presentations, and practical assessments			
Resources Computer or laptop, Python programming environment, OpenCV and face recognition libraries, sample face image dataset, internet access, and development tools (IDE / Jupyter Notebook)			
Qualification of Instructors Instructors shall be qualified in ICT/Computing with knowledge of programming, AI concepts, Python tools, and competency-based training (CBT) methodologies			

LEARNING GUIDE

Module 01	Learning Guide
	Name of technical/vocational training school: Address: Phone No.: Fax No.: E-Mail ID:
Module / Task	Design a Face Recognition Attendance System (AI Solution Design)

Introduction
This task guides the learner to design an AI solution for an attendance system using face recognition. The learner identifies the problem, defines requirements, selects the AI method, and prepares a simple design workflow.

Terminal Performance Objective
Design an AI solution for a Face Recognition Attendance System and explain the solution clearly to the assessor.

Given
Computer or laptop, Python programming environment, OpenCV and face recognition libraries, sample face image dataset, internet access, development tools (IDE / Jupyter Notebook).

You will
Understand the problem clearly; define inputs and expected outputs; design the AI workflow; select suitable tools and methods; prepare a simple design/diagram; present and explain the solution.

How Well
The learner must demonstrate correct understanding of the problem, proper selection of methods and tools, and a clear workflow explanation according to the assessment criteria.

Enabling Objective (EO)
EO1: Identify the problem and user requirements EO2: Define inputs and outputs EO3: Select AI methods and tools EO4: Prepare and explain the solution design

Institution	Dept.	Prog.	Duty	Task	Preqt	Avg. Hrs.	Dates Revised
		AI Developer (NVQ L4)	Support AI System Development	Design Face Recognition Attendance System	Basic ICT knowledge and programming skills	20 Hrs	

Training	Learning Steps	Task 01
EO (1)	1. Understand the purpose of the attendance system 2. Identify users (students, lecturers, admin) 3. Define the problem (manual attendance limitations) 4. Determine system objectives (AI-based attendance)	Design a Face Recognition Attendance System
EO (2)	1. Identify input data (face images via camera/dataset) 2. Define outputs (attendance record with name, date, time) 3. Determine data format and storage method 4. Validate input-output relationship (IPO model)	
EO (3)	1. Select AI method (face recognition) 2. Choose tools (Python, OpenCV, libraries) 3. Identify dataset requirements 4. Plan preprocessing steps 5. Define workflow (capture → process → recognize → mark attendance) 6. Prepare system design	
EO (4)	1. Develop system flowchart/diagram 2. Explain workflow clearly 3. Justify selected tools and methods 4. Present solution to assessor 5. Answer questions confidently 6. Improve solution based on feedback	

Activity 1	INSTRUCTION SHEET	Task B-01
	1. Identify the problem and requirements a) Understand the purpose of the attendance system b) Identify users (students, lecturers, admin) c) Define the problem and system objectives	Design a Face Recognition Attendance System
	2. Define system inputs and outputs a) Identify input data (face images via camera/dataset) b) Define outputs (attendance record: name, date, time) c) Determine data format and storage method d) Validate input-output relationship (IPO model) e) Ensure data accuracy and completeness	
	3. Select methods and design solution a) Select AI method (face recognition) b) Choose tools (Python, OpenCV, libraries) c) Prepare workflow (capture → process → recognize → mark attendance)	

Competency Profile

Element 01 Support AI System Development

A Competent Person Can	A Person Who is not Yet Competent Can
1. Identify problem requirements based on user needs	**1.** Cannot understand or define the problem clearly
2. Define input data and expected outputs	**2.** Unable to identify inputs or expected outputs
3. Select appropriate AI methods (e.g., classification, prediction)	**3.** Chooses incorrect or irrelevant AI techniques
4. Analyze available data sources for suitability	**4.** Does not evaluate data or uses unsuitable data
5. Design a basic solution workflow (input → process → output)	**5.** Cannot organize or explain the workflow
6. Assist in preparing and reviewing design documentation	**6.** Fails to document or improve design properly

Evidence Requirements

Unit of Competency: Assist in Designing AI Solutions	**Candidate's Name:**
Evidence Requirements	**Possible Evidence Gathering Techniques**
1. Identify problem requirements and user needs	1. Observation of learner during task
2. Define inputs and expected outputs	2. Oral questioning / interview
3. Select appropriate AI methods	3. Written test or short answers
4. Analyze and select suitable data sources	4. Review of student notes / report
5. Design basic AI solution workflow (IPO model)	5. Evaluation of flowchart or diagram
6. Prepare and review design documentation	6. Assessment of project/report submission

Evidence Plan

<table>
<tr><td colspan="2">Unit of Competency:
Assist in Designing AI Solutions</td><td>Candidate’s Name:</td></tr>
<tr><th>Sources of Evidence</th><th>Agreed Evidence</th><th>Received</th></tr>
<tr><td>1. Observation of AI solution design process</td><td>1. Direct observation checklist</td><td></td></tr>
<tr><td>2. Student explanation of requirements and workflow</td><td>2. Oral questioning assessment</td><td></td></tr>
<tr><td>3. Design document / report submission</td><td>3. Written report evaluation</td><td></td></tr>
<tr><td>4. Flowchart or system architecture diagram</td><td>4. Diagram review checklist</td><td></td></tr>
<tr><td>5. Dataset identification and analysis notes</td><td>5. Data analysis review</td><td></td></tr>
<tr><td>6. Responses to oral questioning</td><td>6. Structured interview</td><td></td></tr>
<tr><td>7. Project presentation and demonstration</td><td>7. Presentation evaluation rubric</td><td></td></tr>
<tr><td colspan="3">Additional Activity
Identify the problem (automate student attendance)
Define inputs (face images) and outputs (attendance record)
Collect or use a sample face image dataset
Design system workflow (capture → process → recognize → mark attendance)
Select tools (Python, OpenCV, face recognition library)
Explain how the model will recognize faces
Present the solution design to the assessor</td></tr>
<tr><td colspan="3">Arrangements
Provide a computer or laptop for each student
Install required software (Python, OpenCV, face recognition library)
Provide a sample face image dataset
Ensure internet access for resources and support
Prepare development environment (IDE / Jupyter Notebook)
Provide necessary instructions and guidance
Schedule presentation session with assessor
Ensure tools and systems are functioning properly</td></tr>
<tr><td>Submitted On</td><td colspan="2"></td></tr>
<tr><td>Name of Assessor</td><td colspan="2"></td></tr>
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Assessment Instrument Direct Observations Check List

Candidate Name	
Assessor Name	
Unit of Competency	Assist in Designing AI Solutions
Workplace	Computer Lab
Date	
Procedure	Observation of AI solution design

During the performance of skills, did the trainee	Yes	No	N/A
Identify problem requirements	☐	☐	☐
Define inputs and outputs	☐	☐	☐
Select AI method	☐	☐	☐
Design workflow	☐	☐	☐
Explain solution clearly	☐	☐	☐

The candidate’s performance was	Not Satisfactory ☐	Satisfactory ☐
Feedback to candidates		

Trainee Signature	
Assessor Signature	

<u>Assessment instrument checklist for oral questioning</u>

Candidate Name	
Assessor Name	
Unit of Competency	Assist in Designing AI Solutions
Workplace	Computer Lab
Date of assessment	
Procedure	Oral questioning on AI solution design

Oral/Interview Questions	**Satisfactory Response**	
	Yes	**No**
1. What is the purpose of the AI system?	☐	☐
2. What are the inputs and outputs?	☐	☐
3. Which AI method is used?	☐	☐
4. Explain the workflow	☐	☐
5. What tools are required?	☐	☐
The candidate's underpinning knowledge was	Not Satisfactory ☐	Satisfactory ☐

Signed by the Assessor:	Date
Feedback to candidates	
Acceptable answers are	